Language: English

The 3rd International Mathematics Summer Camp (IMSC2025)

Solutions and Marking Schemes-Day I

Problem 1. Let D be a variable point on the interior of side BC of triangle ABC. The circumcircles of triangles ABD and ACD intersect sides AC and AB at $E \neq A$ and $F \neq A$, respectively. The circumcircles of triangles BDF and CDE intersect at $X \neq D$ and the circumcircles of triangles AEF and ABC intersect at $Y \neq A$.

Prove that the line XY passes through a fixed point, independent of the choice of D.

Solution 1. Let M be the midpoint of $[BC]$.

Let line XY intersect (ABC) for the second time at Z.

We see that $\angle XDB = \angle XFA = \angle XYA$. This implies $\angle ACZ = \angle XDC$. Also note that $\angle AZC = \angle ABC = \angle CED = \angle CXD$. Thus the triangles AZC and CXD are similar. Using this, we establish that $\angle XCD = \angle ZAC = \angle ZBC$ i.e. $CX \parallel BZ$. Similarly $BX \parallel CZ$.

Hence $BXCZ$ is a parallelogram. This proves that XY passes through M, as desired. □

Solution 2. The core idea of this solution is to introduce the point $T = (AEFXY) \cap (BXC)$.

We can see that $\angle EXC = \angle EDC = \angle BAC = 180^\circ - \angle FXE$, so X lies on line CF. Similarly X lies on line BE.

Next, we have $\angle TAE = \angle TXE = \angle TCB$. This proves that (ATC) is tangent to line BC. Similarly (ATB) is tangent to line BC. Since M has equal power with respect to (ATB) and (ATC), it follows that A, T, M are collinear. Furthermore, $MT \cdot MA = MB^2 = MC^2$.

Now we consider the inversion of center M and power MB^2. Let Y' be the image of X under this inversion. From the previous remark we get that Y' lies on $(AEFXY)$. We also have that (BXC) is the reflection of (ABC) across BC. In this case, (BXC) is mapped to (ABC), which means that Y' also lies on (ABC). Therefore, Y' lies at the intersection of $(AEFXY)$ and (ABC), implying that $Y' = Y$. Therefore, we have proved that XY passes through M.

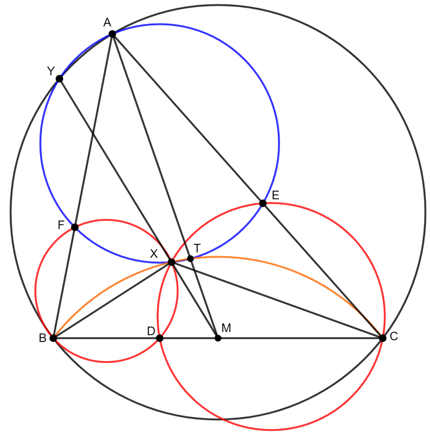

FIGURE 1.

Comment 1. Instead of using the point T in Solution 2, we can prove that the power of M to $(AEFX)$ is MB^2. This simply follows from the fact that the circle of diameter BC is orthogonal to $(AEFX)$, which is a consequence of Brokard Theorem applied to the cyclic quadrilateral $AEFX$ (recall that C lies on the polar of B with respect to $(AEFX)$ if and only if the aforementioned circles are orthogonal).

Solution 3. Define A' to be the reflection of A across BC and S to be the second intersection of line YD with (ABC).

Claim 1: $BXCA'$ is cyclic and A', D, X are collinear.

Proof. We have $\angle BXD = \angle BFD = \angle ACB$ and similarly $\angle CXD = \angle ABC$. Then $\angle BXC = 180^\circ - \angle BAC = 180^\circ - \angle BA'C$, which proves the first result. For the second one, notice that $\angle BXA' = \angle BCA' = \angle ACB = \angle BXD$.

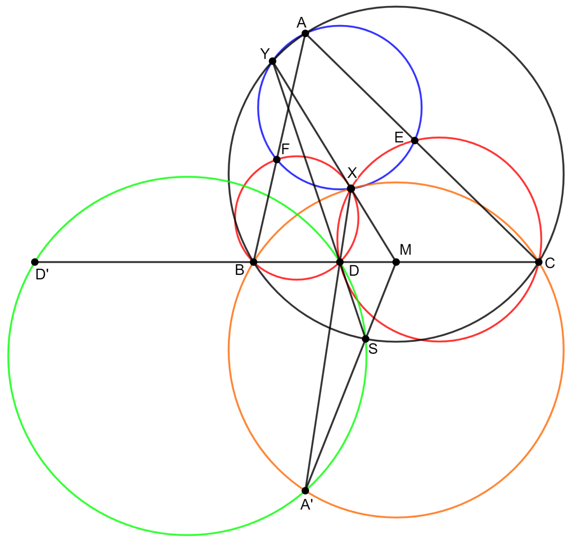

FIGURE 2.

Claim 2: $\frac{BS}{CS} = \frac{AB}{AC}$.

Proof. By angle-chasing we get that triangles BYF and CYE are similar, so $\frac{BY}{CY} = \frac{BF}{CE}$. Also, power of point gives the identities $BD \cdot BC = BF \cdot BA$ and $CD \cdot BC = CE \cdot AC$. Thus we can write

$$\frac{BS}{CS} \cdot \frac{BY}{CY} = \frac{BY}{CY} \cdot \frac{\sin\angle BYD}{\sin\angle CYD} = \frac{[BYD]}{[CYD]} = \frac{BD}{CD} = \frac{BF \cdot BA}{CE \cdot CA} = \frac{BY}{CY} \cdot \frac{BA}{CA},$$

where the first identity follows from sine rule in (ABC).

This implies that $\frac{BS}{CS} = \frac{BA}{CA}$, as desired.

From the second claim it follows that the quadrilateral $ABSC$ is harmonic. It is well-known that S and A' are swapped by an inversion of center M and power MB^2.

Introduce the point D' such that $(D, D'; B, C) = -1$. By well-known properties of harmonic bundles, we have $MD \cdot MD' = MB^2$ and $DD' \cdot MD = BD \cdot CD$. From the first relation we get $MD \cdot MD' = MS \cdot MA'$, so $DD'A'S$ is cyclic.

To finish the problem, consider an inversion of center D and power $-BD \cdot CD$. This inversion maps A', D' and S to X, M, and Y respectively. As $DD'A'S$ is cyclic, it follows that X, Y, M are collinear, as desired.

Comment 2. We have that X is the $Y-$Humpty point of triangle BYC. This is immediate to see in the first two solutions. It's also possible to prove this fact without introducing additional points: angle-chase to get $\angle XYC = \angle XCB$ and $\angle XYB = \angle XBC$, which prove that (YXC) and (YXB) are tangent to BC i.e. X is the Humpty point. In fact, the previous angle relations are equivalent to $BXCZ$ being a parallelogram in the first solution.

Solution 4: Let line XY intersect (ABC) for the second time at Z.

We see that $\angle ABZ = \angle AYZ = \angle AYX = \angle AFX$, implying $BZ \parallel CF$.

On the other hand, we have $\angle EXC = \angle EDC = \angle BAC = 180° - \angle FXE$, so point X lies on the line CF.

Hence lines CX and BZ are parallel, and similarly lines BX and CZ are parallel.

We conclude that $BXCZ$ is a parallelogram, proving that XY passes through M, as claimed. $\square$

Marking Schemes:

- **Solution 1:**
  - Claiming M (midpoint of BC) is on XY - 1 point
  - $\Delta AZC \sim \Delta CXD$ - 2 points
  - $CX \parallel BZ$ and $BX \parallel CZ$ - 2+1 points
  - Conclusion - 1 point
- **Solution 2:**
  - Claiming M (midpoint of BC) is on XY - 1 point
  - $\overline{X, C, F}$ and $\overline{X, B, E}$ - 1+1 points
  - $(ATB), (ATC)$ tangent to BC - 1+1 points
  - $M \in AT$ - 1 point
  - Inversion/ conclusion - 1 point

- **Solution 3:**
  - Claiming M (midpoint of BC) is on XY - 1 point
  - $BXCA'$ cyclic - 1 point
  - $\overline{A', D, X}$ - 1 point
  - $ABSC$ harmonic quadrilateral - 1 point
  -$\Delta BYF \sim \Delta CYE$ - 1 point
  - All the metric relations needed for inversion - 1 point
  - Inversion/ conclusion - 1 point
- **Solution 4:**
  - Claiming M (midpoint of BC) is on XY - 1 point
  - $\overline{X, C, F}$ and $\overline{X, B, E}$ - 1+1 points
  - $CF \parallel BZ$ and $BE \parallel CZ$ - 2+1 points
  - Conclusion - 1 points
- **Remark:** The points awarded within different solutions are not additive.

Problem 2. Let $n \geqslant 3$ be a positive integer and let $(x_1, x_2, \ldots, x_n)$ be an n-tuple of nonnegative real numbers such that $x_1 + x_2 + \cdots + x_n = 1$. Find, over all such n-tuples, the smallest possible value of

$$\max\left\{x_i + \sqrt{x_{i+1}x_{i+2}} \mid i = 1, 2, \ldots, n\right\}.$$

(Here we take $x_{n+1} = x_1$ and $x_{n+2} = x_2$.)

Solution. The answer is $\frac{1}{\lfloor 2n/3 \rfloor}$ for $n > 1$, and (trivially) 2 for $n = 1$.

Denote the considered maximum by M. We observe that $x_i + \min\{x_{i+1}, x_{i+2}\} \le M$. Together with $\max\{x_{i+1}, x_{i+2}\} \le M$ this implies that

$$x_i + x_{i+1} + x_{i+2} \le 2M.$$

It follows that $x_i + x_{i+1} + \cdots + x_{i+3k-1} \le 2kM$ for all i and k.

- If $n = 3k$ ($k \ge 1$), then $1 = x_1 + \cdots + x_{3k} \le 2kM$, so $M \ge \frac{1}{2k}$.
- Let $n = 3k + 2$. Choose i such that $x_{i+1} \le x_{i+2}$; as above, this implies that $x_i + x_{i+1} \le M$. Moreover $x_{i+2} + x_{i+3} + \cdots + x_{i+3k+1} \le 2kM$. Summation yields $1 \le (2k+1)M$, so $M \ge \frac{1}{2k+1}$.
- Let $n = 3k + 1$ ($k \ge 1$). We claim that $x_j + x_{j+1} + x_{j+2} + x_{j+3} \le 2M$ holds for at least one index j. Consider i such that $x_{i+1} \ge x_{i+2}$; then, as above, $x_i + x_{i+2} \le M$. If any of the inequalities $x_{i-1} + x_{i+1} \le M$ and $x_{i+1} + x_{i+3} \le M$ is true, we can take $j = i - 1$ or $j = i$. Else the opposite inequalities $x_{i-1} + x_i \le M$ and $x_{i+1} + x_{i+2} \le M$ hold, so we can take $j = i - 1$.

  Moreover, $x_{j+3} + x_{j+4} + \cdots + x_{j+3k-1} \le 2(k-1)M$. Summation yields $x_1 + x_2 + \cdots + x_n \le 2kM$, so $M \ge \frac{1}{2k}$.

Therefore, for $n > 1$ we always have $M \ge \frac{1}{\lfloor 2n/3 \rfloor}$. This value is attained if $x_i = 0$ for $i \equiv 1 \pmod 3$ and $x_i = M$ otherwise. □

Marking Scheme:

- $x_i + x_{i+1} + x_{i+2} \le 2M$ $\quad -$ 1 point
- Proof for $n = 3k$ $\quad -$ 1 point
- Proof for $n = 3k + 2$ $\quad -$ 1 point
- Proof for $n = 3k + 1$ $\quad -$ 2 points
- Construction of all 3 cases, with the correct answer $-$1 point

If a student manages to get 6 points from the above, award 1 point for completeness.

Problem 3. Alice and Bob play a game on a K_{2026}. They take turns with Alice playing first. Initially all edges are uncoloured. On Alice's turn she chooses any uncoloured edge and colours it red. On Bob's turn, he chooses $1, 2$ or 3 uncoloured edges and colours them blue. The game ends once all edges have been coloured. Let r and b be the number of vertices of the largest red and blue clique, respectively. Bob wins if at the end of the game $b > r$. Prove that Bob has a winning strategy.

Note: K_n denotes the complete graph with n vertices and where there is an edge between any pair of vertices. A red (blue, respectively) clique refers to a complete subgraph of which all the edges are red (blue, respectively).

Solution:

We will present some edges Bob has to colour in every situation.

At the start of each turn of Alice, we assume that an even number $2k$ of vertices have at least one incident coloured edge, and Bob has paired these in $k \leq 1013$ pairs of vertices (v_i, v_i'), which we can consider as diametrically opposite, and possibly one special edge xy (mentioned later). Hereby all edges $v_i v_i'$ are coloured by Bob, and if $v_i v_j$ is coloured by Alice, then $v_i' v_j'$ is coloured by Bob (and either 0 or 2 edges in $G[v_i, v_i', v_j', v_j]$ are uncoloured). Here one can observe that $(v_j')' = v_j$ (so in further steps, one can use v_j for the vertex called v_j' before).

If Alice colours an edge between two new vertices v_i, v_j, Bob takes two new ones as well (if possible), calls them v_i', v_j', and colours the edges $v_j v_j', v_i v_i'$ and $v_i' v_j'$.

If Alice colours an edge containing one labelled v_i and one unlabelled vertex (call it v_j), Bob chooses an unlabelled vertex as well, calls it v_j', and colours the edges $v_j v_j'$ and $v_i' v_j'$.

If Alice colours an uncoloured $v_i v_j$ (both vertices labelled), Bob colours the remaining (1 or 3) edges on the K_4 spanned by $\{v_i, v_j, v_i', v_j'\}$.

If the latter was impossible, Bob calls this special edge coloured by Alice xy, and colours one edge between y and a vertex different from x and y. Whenever $v_i x$ (or $v_i y$) is coloured by Alice, Bob colours all remaining edges in the K_4 induced by x, y, v_i, v_i'. (there are always 3, except if one of its edges was coloured in the turn immediately after Alice coloured xy)

At the end, Alice has no clique with at least 3 vertices coloured containing xy, while Bob clearly has created blue K_3s.

If the largest red clique of Alice contains $\{v_i \mid i \in I\}$, then Bob has a blue clique on $\{v_i' \mid i \in I\}$. Furthermore, there are at most $|I| - 1$ many edges of the form $v_i v_j'$ with $i, j \in I$ coloured by Alice, since a K_4 spanned by vertices of the form $v_i v_j v_i' v_j'$ can only contain two blue edges if Alice coloured one of these edges choosing at least one previous unlabelled vertex. Putting the pairs in order (equivalently the indices of I), an additional blue edge only appears starting from the second vertex.

If the largest red clique of Alice contains $\{v_i \mid i \in I\} \cup \{x\}$, then Bob has a blue clique on $\{v_i' \mid i \in I\} \cup \{v_j\}$ as before, which can be extended by adding y to a larger blue clique.

We conclude that with the above strategy, at the end of the game, the largest blue clique is always larger than the largest red clique.

Marking Scheme: Additive points for the main solution

- Idea of pairing vertices; where Bob will claim the edges vv' and u'v' whenever Alice claims uv - 1 point
- *For each of the 4 cases - 1 point* (if reaction involves 1 to 3 edges)
  - how to act on 2 new vertices - 1 point
  - how to act on old + new vertex - 1 point
  - how to act on 2 old vertices - 1 point
  - how to act on xy and old vertex and x or y - 1 point
- final conclusion - 2 points

General (additive):

- giving working strategy - 5 points
- proving strategy works - 2 points

Remarks plausible alternatives

For not necessarily extendable solutions (but complete and correct) with more edges (non-additive with anything else)

- 4 edges - 3 points
- 5 edges - 2 points
- 6 till 16 edges - 1 point
- rigorous maker breaker idea without explanation for small n - 1 point
- rigorous maker breaker idea with convincing sketch for n=2026 - 2 points

General remarks

- proving $b \geq r$: 0 point
- minor flaw: -1 point
- major flaw: -2 points
- The points awarded within different solutions are not additive.
- 2 edges (extendable to all n) - special award